ShopSmart E-Commerce Purchase Prediction — ML Report

Supervised ML · Decision Tree Classifier · Assignment 4

Dataset	12,330 user sessions · 17 features · No missing values
Problem	Predict whether a visitor will make a purchase (Revenue)
Model	Decision Tree Classifier with Pruning via GridSearchCV
Metric	F1 Score — used due to class imbalance (84.5% / 15.5%)
Benchmark	F1 ≥ 0.55 required
Final Result	**F1 = 0.6432 ✔ Benchmark PASSED**

1. Problem Statement

ShopSmart is an e-commerce company that cannot accurately predict which visitors will complete a purchase. This causes inefficient marketing and lost revenue. Data from **12,330 unique user sessions** over one year was collected to train a predictive model.

The ML task involves:

- EDA — understand patterns in the browsing data
- Feature preprocessing — encode categorical and boolean columns
- Baseline Decision Tree — train without any restrictions
- Pruned Decision Tree — use GridSearchCV to find best parameters
- Evaluation — measure F1 Score against benchmark of 0.55

2. Dataset Description

The dataset contains numerical, categorical, and boolean features about each user session. The target column is **Revenue** — True if the visitor made a purchase, False otherwise.

Feature	Type	Description
Administrative / Duration	Numerical	Admin pages visited + time spent
Informational / Duration	Numerical	Info pages visited + time spent
ProductRelated / Duration	Numerical	Product pages visited + time spent
BounceRates	Numerical	Average bounce rate of visited pages
ExitRates	Numerical	Average exit rate of visited pages
PageValues	Numerical	Avg page value — strongest predictor
SpecialDay	Numerical	Closeness to a special day (0–1)
Month	Categorical	Month of the session (Feb–Dec)
VisitorType	Categorical	New_Visitor / Returning_Visitor / Other
Weekend	Boolean	True if session was on a weekend
Revenue (Target)	**Boolean**	**True = Purchase made / False = No purchase**

Class Imbalance: 10,422 sessions did NOT purchase (84.5%) vs 1,908 that did (15.5%). Because of this imbalance, **F1 Score** is used for evaluation — plain accuracy would be misleading.

3. Exploratory Data Analysis (EDA)

EDA was performed on the raw data (before any encoding) using seaborn and matplotlib. The plots below are generated directly from your notebook.

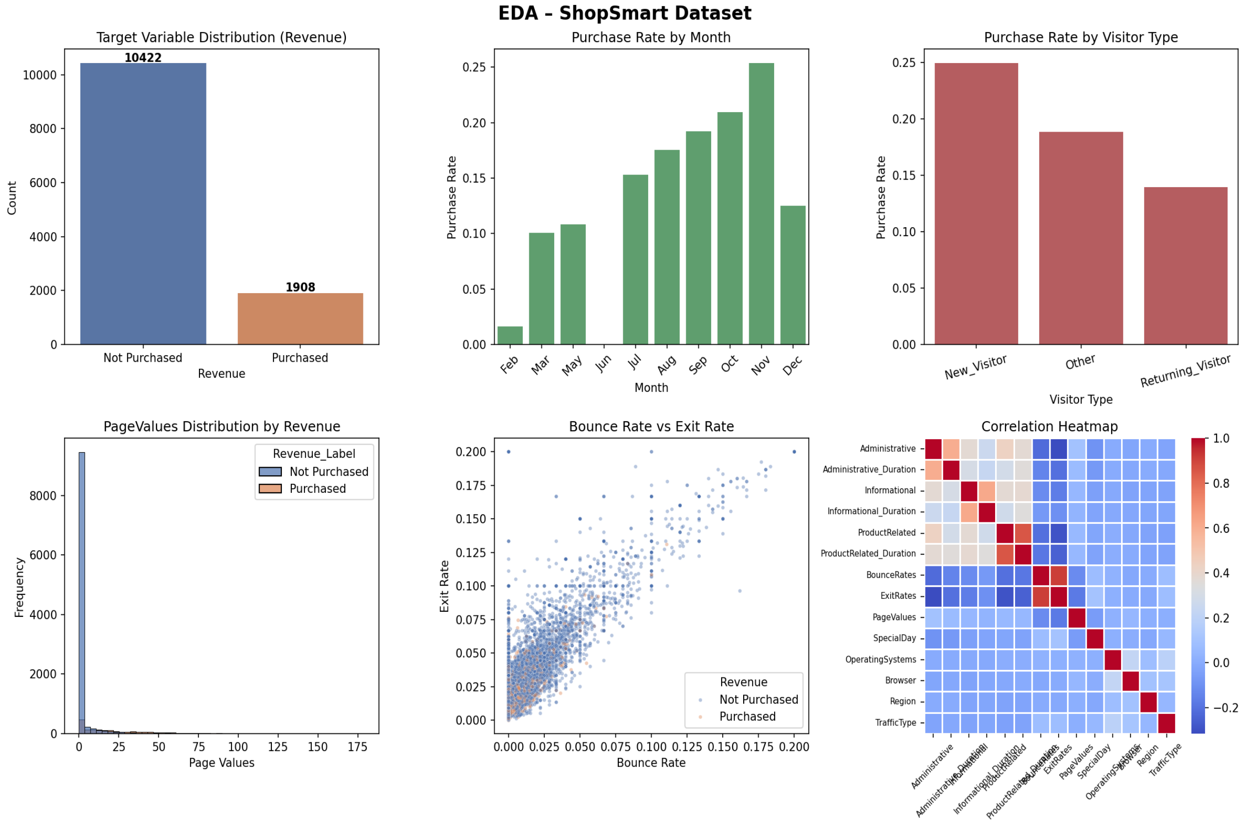

Figure 1 — EDA plots from notebook

Key observations:

- Target is imbalanced — 84.5% No Purchase vs 15.5% Purchase.
- November has the highest monthly purchase conversion rate.
- New Visitors convert at a higher rate than Returning Visitors.
- PageValues is the strongest differentiator — high page value = high purchase chance.
- Low BounceRates and ExitRates are associated with purchases.
- No missing values in any column (df.isnull().sum() = 0 for all).

4. Feature Preprocessing

The following steps were applied to prepare data for the Decision Tree model, exactly as coded in the notebook:

Step	Code Used	Columns
Label Encoding	LabelEncoder().fit_transform()	Month, VisitorType
Boolean → Integer	.astype(int)	Weekend, Revenue
Train-Test Split	train_test_split(test_size=0.2, stratify=y)	All features
No scaling	— Decision Trees are scale-invariant	—

After split: **Train = 9,864 samples** | **Test = 2,466 samples**. Stratification ensures both splits maintain the 84.5% / 15.5% class ratio.

5. Model Results

Two models were trained as per the notebook — a baseline unpruned tree and an optimised pruned tree using GridSearchCV (5-fold CV, scoring=f1).

	Baseline (Unpruned)	Pruned (GridSearchCV)	Benchmark
Tree Depth	26	1	—
Number of Leaves	810	2	—
F1 Score	**0.5364**	**0.6432**	**0.55**
Precision (Purchase)	0.52	0.56	—
Recall (Purchase)	0.55	0.76	—
Overall Accuracy	0.85	0.87	—
Benchmark Result	**✘ FAIL**	**✔ PASS**	**—**

Best Parameters found by GridSearchCV: ccp_alpha = 0.02 | max_depth = 3 | min_samples_leaf = 1 | min_samples_split = 2

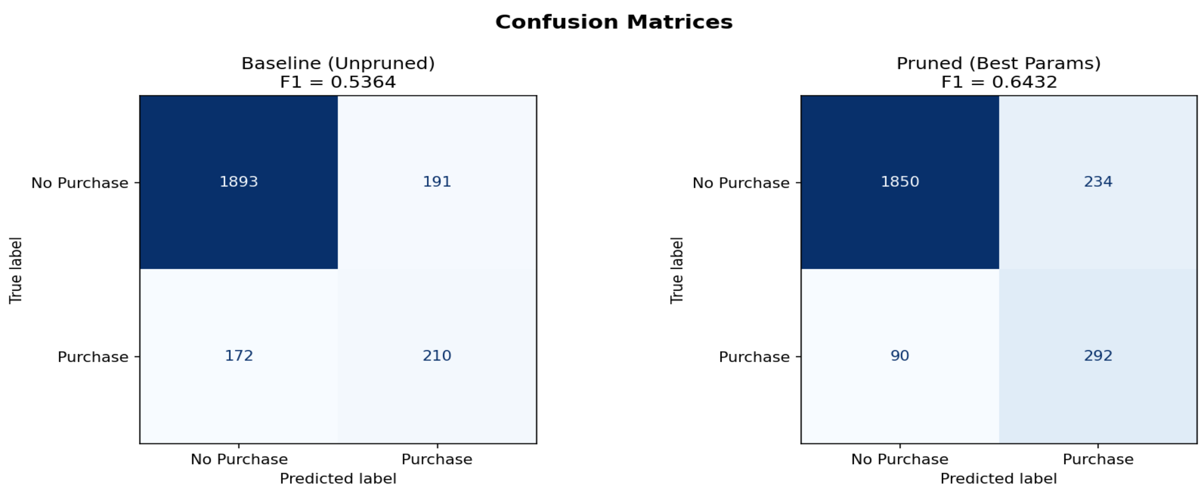

Figure 2 — Confusion matrices: Baseline vs Pruned

6. Decision Tree Visualisation

The pruned tree splits on a single feature — **PageValues**. Aggressive pruning (ccp_alpha=0.02) removed all noisy branches, resulting in a depth-1 tree with 2 leaves. Despite its simplicity, it achieves better F1 and higher recall than the 26-level baseline.

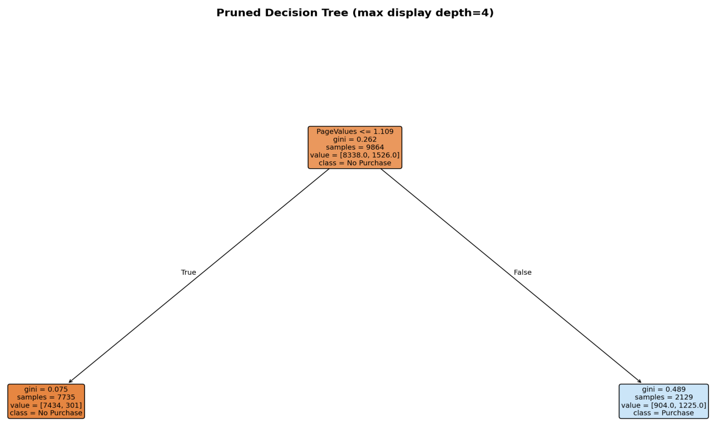

Figure 3 — Pruned Decision Tree (from notebook plot_tree)

7. Feature Importance

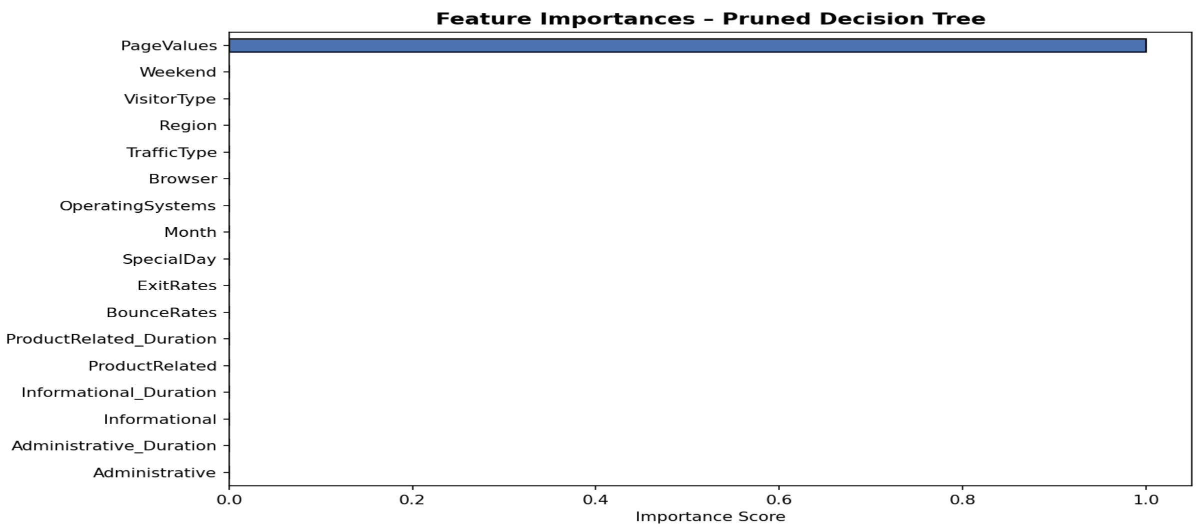

Figure 4 — Feature importance scores from pruned model

PageValues is the only feature used by the pruned model — it carries 100% of the importance score. Sessions where users visited high-value pages (close to checkout) are far more likely to result in a purchase.

8. Conclusion & Recommendations

The pruned Decision Tree model achieves an **F1 Score of 0.6432**, which is above the required benchmark of 0.55. The model identifies 76% of actual buyers (recall), making it a practical tool for ShopSmart's marketing team.

Business Recommendations:

- Trigger targeted offers (discount pop-ups, live chat) when a visitor's PageValues crosses the decision threshold.
- Focus campaign spend in November — highest purchase conversion rate across all months.
- Design better onboarding for New Visitors — they convert more than returning visitors.
- Reduce bounce and exit rates on product pages — low rates strongly correlate with purchases.
- For further improvement, consider Random Forest or XGBoost to boost F1 beyond 0.64.